

DREAM: Document Recognition with Explicit Adaptive Memory

Tianqi Zhao¹, Di Wu¹, Liangrui Peng^{1,*}, Yifan Huang¹, Kemeng Zhao¹,
 Shuo Li¹, Zhiyu Li¹, Yizhu Wang¹, Borui Jiang², Yuyang Li²

¹Tsinghua University

²Huawei Noah's Ark Lab

ztq24@mails.tsinghua.edu.cn

Abstract

Large multimodal models (LMMs) have shown promising performance for various document recognition tasks. However, LMMs adopt implicit modeling, and the parameters lack interpretability. Inspired by recent advances in human memory and learning research, we propose an **explicit multiscale prototype memory** that augments document recognition models, explicitly modeling recurrent layout and stylistic patterns across different spatial resolutions. A Memory Retrieval Mechanism enables local regions to sparsely attend to a few prototypes (e.g., image borders, tilted text); the retrieved compositional factors are concatenated with visual features and passed to the decoder, providing explicit region-wise structural context. Prototype memory consolidation updates and stabilizes prototypes via attention-weighted exponential moving average (EMA) strategy, while sparsity and anti-collapse regularization promote selective activation. We further adopt hierarchical memory for multi-resolution encoding. The proposed DREAM module is a plug-and-play component, allowing seamless integration into various encoder-decoder architectures. We validate on two tasks including document recognition on public datasets and the self-built DreamDoc dataset, and handwriting recognition on the SCUT-HCCDoc and SCUT-EPT datasets. Experimental results show that the proposed method is effective. The DreamDoc dataset and main code are available at github.com/TianqiZhao-THU/DREAM.

1. Introduction

Large multimodal models (LMMs) have recently achieved strong performance on document parsing and recognition, thanks to unified modeling of text, layout, and visual signals. [7, 8, 18, 24, 27, 38, 39].

However, existing LMM-based document recognition methods are largely black-box parametric models: their knowledge is implicitly stored in network weights, making it

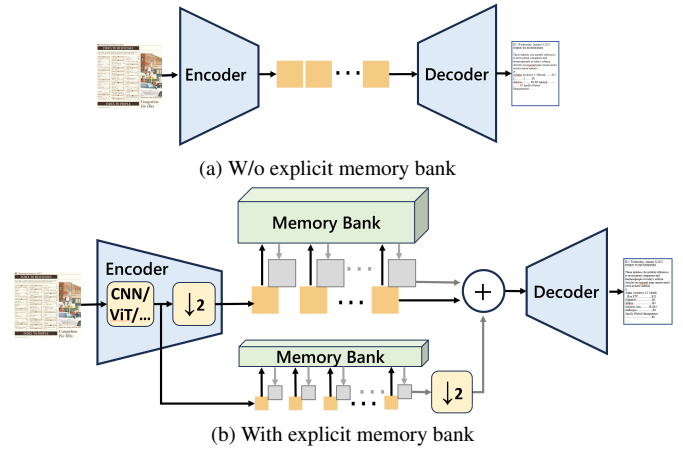


Figure 1. Illustrations of different frameworks (a) Encoder-decoder framework w/o explicit memory (b) the propose DREAM

difficult to disentangle which spatial or stylistic factors contribute to recognition. The visual representations obtained from document regions are often entangled with complicated factors of spatial structure and visual styles. Without an explicit mechanism to factorize these components, models struggle to maintain robustness under complex or unseen layouts. Moreover, current methods are purely parametric: all knowledge is stored in model weights, so their representational capacity is directly tied to parameter count, even when abundant training data is available. Furthermore, these methods have difficulty in extension and updating. When encountering new styles or domain-specific documents, such models typically require costly parameter updates (e.g., fine-tuning) for adaptation, and lacking a flexible non-parametric memory mechanism that would allow knowledge to be expanded rapidly.

Real-world documents exhibit rich multi-scale variability: global page layouts (e.g., headers/footers, multi-column organization), structured mid-level components (figures, formula regions), and fine-grained visual styles (font changes, skew, noise). In addition, inconsistent capture and scanning

* Correspondence to: Liangrui Peng, penglr@tsinghua.edu.cn

resolutions amplify cross-domain appearance shifts. As a result, models that rely solely on parametric representations struggle to disentangle these compositional factors.

To bridge these gaps, we introduce an explicit, adaptive, and multiscale prototype memory that augments document recognition models with non-parametric structural knowledge. To the best of our knowledge, we are the first to introduce prototype memory to document recognition. Our key insight is that documents, unlike natural scene images, are governed by a finite set of recurring structural patterns. This inherent regularity makes them highly suitable for prototype memory clustering. Our memory module learns a set of prototypes that summarize these frequently occurring layout structures and stylistic factors across the training corpus.

For instance, when a patch contains both the boundary of an embedded image and a segment of slanted text, the prototype memory mainly activates two slots “image borders” and “tilted text,” and returns their encoded patterns as auxiliary representations. These representations are fused with the visual encoder output and forwarded to the decoder.

We mathematically prove that the posterior distribution of latent variables in a Gaussian Mixture Model (GMM) can be exactly represented by a learnable cross-attention [35] mechanism. Therefore, our prototype memory employs an attention-based read-write mechanism: regional features serve as queries, while prototype vectors function as keys and values for differentiable addressing. During training, the model uses attention-based reading to retrieve the most relevant compositional information for each region, while a momentum-based exponential moving averages (EMA) writing mechanism updates the prototypes continuously according to attention weights, allowing them to aggregate frequently occurring structural and stylistic patterns.

Furthermore, we instantiate the prototype memory as a multiscale hierarchy: encoder features at different spatial resolutions access separate prototype banks, and the retrieved representations are converted to a common space and fused.

To encourage each region to activate only a small number of prototypes, we apply sparsity and anti-collapse regularization to the attention distributions. During inference, the memory operates in a read-only snapshot mode. Because attention weights explicitly indicate which prototypes are activated, the entire mechanism is naturally interpretable.

The proposed explicit adaptive multiscale memory module can be seamlessly integrated into a wide range of encoder-decoder architectures in a plug-and-play manner. We apply our module in two models: a document recognition model and a handwriting text line recognition model. For document recognition, we train on a self-constructed DreamDoc dataset, and the Vary-600K dataset [36], and evaluate on Fox [25] and DreamDoc. For handwriting text line recognition, we conduct experiments on SCUT-HCCDoc [43] and SCUT-EPT [45]. Our results demonstrate that the proposed

memory module consistently improves recognition performance and robustness under diverse layouts and styles.

Our main contributions are summarized as follows:

- We propose an **explicit multiscale prototype memory** that leverages nonparametric structural knowledge to enhance document recognition models and implements a prototype memory consolidation and retrieval mechanism to explicitly model recurrent layout and stylistic patterns across different spatial resolutions.
- We introduce an **entropy-based sparsity regularization** on attention distributions to encourage sparse and non-collapsing prototype assignments.
- We implement a **plug-and-play** module, DREAM, and integrate it into both a large multimodal document recognition model and a handwriting text line recognition model. Experiments are conducted on public and self-built datasets to validate the proposed methods.

2. Related Work

2.1. Document Recognition

Recent advances in document understanding have leveraged multi-modal architectures to capture both visual and textual information. While the semantic interpretation of document structures is often addressed through multi-stage layout analysis pipelines (e.g., LayoutParser [32], SwinDocSegmenter [2]) and supported by layout-centric datasets like PubLayNet [44] and DocLayNet [30], subsequent approaches have sought more integrated representations. LayoutLM [39] and its variants jointly learn text and layout representations within a unified framework for document-level pre-training. Donut [18] adopts an end-to-end transformer-based approach that eliminates the need for OCR pre-processing in visual document understanding. More recently, large vision-language models have demonstrated promising capabilities: GOT [38] employs high-resolution encoders for fine-grained recognition of diverse optical signals, including plain text, mathematical and molecular formulas, tables, charts, sheet music, and geometric shapes, while supporting multiple output formats; Monkey [24] introduces high-resolution processing that enables more detailed visual capture and enhances comprehensive scene descriptions; InternVL [7] scales up vision-language foundation model pretraining; DeepSeek-VL [27] exhibits strong zero-shot generalization across diverse document types; and PaddleOCR-VL [8] introduces a compact vision-language model for multilingual document parsing. However, these approaches rely primarily on parametric knowledge stored in model weights, lacking explicit mechanisms to retrieve and utilize learned layout patterns. Our method introduces a hierarchical memory module that explicitly stores and retrieves distinctive structural patterns, enabling better adaptation to diverse document layouts without extensive fine-tuning.

2.2. Memory Networks

Biological memory relies on coordinated interactions between the prefrontal cortex (PFC) and hippocampus (HPC) [16], i.e., the hippocampus acts as a sparse, pattern-separated system for rapidly learning episodic memories, and the neocortex functions as a distributed, overlapping system for gradually integrating across episodes to extract latent semantic structure [28]. Jonathan et al. [9] showed that PFC oscillations dynamically regulate hippocampal activity, and that the HPC contains category-selective neurons. Recent work further demonstrates a plasticity-dependent competition rule within the PFC-HPC circuit, whereby neuronal ensembles compete for representational dominance, with the winning ensemble forming the memory trace [17, 34].

Memory mechanisms have been extensively explored to enhance neural networks' capacity and adaptability. Early frameworks such as Neural Turing Machines [10] and Differentiable Neural Computers [11] introduce external memory modules with attention-based read and write operations, enabling models to store and retrieve information beyond their fixed parameters.

In vision tasks, memory networks have been applied to video understanding [12], where temporal dependencies are modeled through structured memory banks. These systems typically use predictive attention over compressed memory sets, enabling future states to be synthesized as convex combinations of condensed representations and thereby supporting efficient multi-hypothesis generation. MovieChat+ [33] further employs Transformer tokens as memory carriers and introduces a dedicated memory mechanism to reduce the computational and storage costs of maintaining long-term temporal dependencies in long-video understanding.

Prototype memory employs a compact set of interpretable prototype vectors to summarize the semantic patterns of a class or a set of subclasses. These prototypes compress large amounts of high-dimensional sample features into a set of retrievable memory slots, which are typically stored in a Key-Value format. During training, prototype memory is commonly updated via Exponential Moving Averages (EMA) or clustering-based assignment [21], allowing incoming sample features to be allocated and written into the corresponding prototypes. During inference, the model accesses this memory through similarity-based [20] or attention-based [4] read-write mechanism.

Most existing memory mechanisms in vision tasks operate within a single sample. They help capture local contextual information but do not enable cross-sample information flow. In document recognition, where complex layouts and diverse font styles pose substantial challenges, we introduce a corpus-level memory mechanism. Inspired by biological memory, our prototype memory mirrors the functional division of the PFC-HPC circuit: the former modulates memory writing and retrieval, while the latter provides category-

structured storage. In addition, we adopt a sparse activation strategy to emulating the competitive writing process observed in biological memory systems.

3. Methods

3.1. Explicit Adaptive Prototype Memory

3.1.1. Prototype Memory as Centers of GMM

In document images, we model the feature token $\mathbf{x}$ extracted from a local region as a weighted combination of several Gaussian sub-components, i.e., a Gaussian Mixture Model (GMM):

$$p(\mathbf{x}) = \sum_{m=1}^M \pi_m \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_m, \boldsymbol{\Sigma}_m) \quad (1)$$

These components include not only the semantic representation of the text itself, but also additive information that co-occurs with semantics, such as complicated and multi-scale layout structures (headers/footers, multi-column layouts, tables, image-text mixtures, etc.) and various stylistic factors (skew, blur, font-size variation, edge transitions, and so on).

To explicitly model such additive information, we introduce a prototype memory that compresses frequently occurring structural or stylistic patterns in the training data into a set of retrievable prototype vectors. Each prototype is represented by the mean $\boldsymbol{\mu}_m$ of a sub-distribution in Eq. 1.

3.1.2. Memory Retrieval Mechanism

The process of retrieving from the prototype memory can be interpreted as estimating the linear combination weight of each prototype component on the input feature. We refer to this estimation as the responsibility of each prototype. Numerically, they are equal to the posterior probability that the latent variable z equals m in Eq. 1:

$$r_m(\mathbf{x}) \triangleq p(z = m | \mathbf{x}) = \frac{\pi_m \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_m, \boldsymbol{\Sigma}_m)}{\sum_{i=1}^M \pi_i \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)} \quad (2)$$

Here, $r_m(\mathbf{x})$ denotes the responsibility of prototype m for the feature $\mathbf{x}$. The latent variable $z \in \{1, \dots, M\}$ indicates which Gaussian component generated the observed feature, and π_m denotes the prior probability of prototype m , reflecting how frequently each additive pattern appears.

We approximate each covariance matrix with an isotropic form $\sigma^2 \mathbf{I}$, where $\mathbf{I}$ is the identity matrix. Taking the logarithm of each term in Eq. 2, we define the scoring function $s_m(\mathbf{x})$:

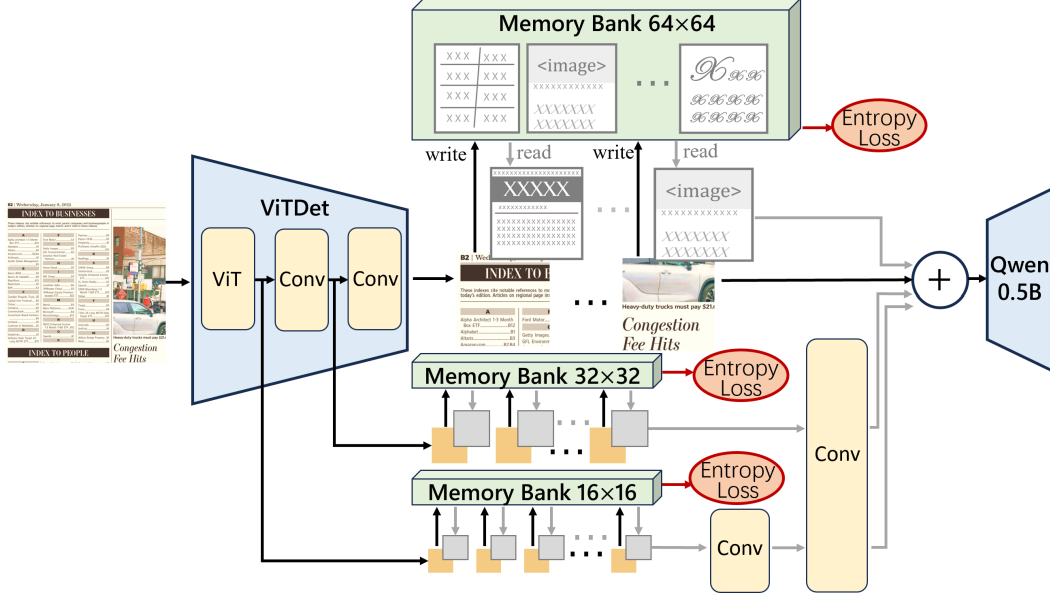


Figure 2. DREAM: Document Recognition model with Explicit Adaptive Multiscale Memory. Visual features with different resolutions serve as queries to access the corresponding memory banks (black arrows). The read operation retrieves prototype memories (gray arrows), which are concatenated with the encoder’s output and fed into the decoder. Each memory bank incorporates an entropy-based sparsity loss.

$$\begin{aligned}
 s_m(\mathbf{x}) &= \log(\pi_m \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_m, \sigma^2 \mathbf{I})) \\
 &= \log \pi_m - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_m)^\top (\sigma^2 \mathbf{I})^{-1} (\mathbf{x} - \boldsymbol{\mu}_m) - \frac{D}{2} \log(2\pi\sigma^2) \\
 &= \frac{1}{\sigma^2} \boldsymbol{\mu}_m^\top \mathbf{x} - \frac{1}{2\sigma^2} \|\boldsymbol{\mu}_m\|^2 + \log \pi_m + C
 \end{aligned} \quad (3)$$

where C denotes a term independent of m . The responsibilities thus take a softmax form:

$$r_m(\mathbf{x}) = \frac{\exp(s_m(\mathbf{x}))}{\sum_{j=1}^M \exp(s_j(\mathbf{x}))} = \text{Softmax}_m(s(\mathbf{x})) \quad (4)$$

This shows that we can compute responsibilities in the GMM using an attention mechanism. The cross-attention mechanism [35] with bias is given by:

$$\text{CA}(\mathbf{q}, \mathbf{k}, \mathbf{v}) = \text{Softmax}\left(\frac{\mathbf{q}W_q W_k^\top \mathbf{k}^\top}{\sqrt{D}} + B\right) (\mathbf{v}W_v) \quad (5)$$

where $\mathbf{q}$ denotes the query, $\mathbf{k}$ the key, $\mathbf{v}$ the value, and B is a bias term.

We take the regional feature (representing the visual modality) $\mathbf{x}$ as the query, i.e., $\mathbf{q} = \mathbf{x}$, and use each prototype mean vector (representing the structural modality) as both key and value, i.e., $\mathbf{k}_m = \boldsymbol{\mu}_m$ and $\mathbf{v}_m = \boldsymbol{\mu}_m$. Substituting these into Eq. 5, the attention weight corresponding to prototype m is:

$$\alpha_m(\mathbf{x}) = \text{Softmax}\left(\frac{\mathbf{x}^\top W_q^\top W_k \boldsymbol{\mu}_m}{\sqrt{D}} + B\right) \quad (6)$$

Comparing this with the responsibility expression in Eq. 4, when there exist learnable projection matrices W_q, W_k and bias B such that the attention weights share the same structure as responsibilities.

$$\frac{\mathbf{x}^\top W_q^\top W_k \boldsymbol{\mu}_m}{\sqrt{D}} \approx \frac{1}{\sigma^2} \mathbf{x}^\top \boldsymbol{\mu}_m, B_m \approx \log \pi_m - \frac{1}{2\sigma^2} \|\boldsymbol{\mu}_m\|^2 \quad (7)$$

Thus, the attention weight $\alpha_m(\mathbf{x})$ can be viewed as a learnable approximation to the responsibility $r_m(\mathbf{x})$.

In Eq. 5, the cross-attention output $M(\mathbf{x})$ is the linear combination of all prototype vectors weighted by their attention weights. The retrieved memory is concatenated with the original regional feature, and the resulting visual representation is projected into the language embedding space before being input to the decoder:

$$\tilde{\mathbf{x}} = \text{Proj}[\mathbf{x} \oplus M(\mathbf{x})] \quad (8)$$

where Proj denotes the projection operation and $\oplus$ denotes the concat operation.

3.1.3. Prototype Memory Consolidation

During the training stage, the prototype memory is not only retrieved but also continuously consolidated, allowing it to

dynamically aggregate frequently occurring patterns from training samples. Therefore, we require the update process to satisfy the following two properties:

Differentiable addressing: each token produced by the encoder should automatically select its write target according to its responsibility with respect to the prototypes. Smooth update: the memory should be updated gradually, in order to avoid catastrophic forgetting.

Based on these considerations, we employ the same attention mechanism used in the reading stage to perform memory routing. For an input feature $\mathbf{x}$, we compute its attention weights $\alpha_m(\mathbf{x})$ with respect to each prototype μ_m using Eq. 6, which correspond to the estimated responsibility $r_m(\mathbf{x})$ defined in Sec. 3.1.1.

Given a batch of image features $\{\mathbf{x}_{b,n}\}$ (where $b = 1, \dots, B$ denotes the batch index and $n = 1, \dots, N$ the spatial position within each sample), the value features are aggregated according to their attention weights to obtain the writing increment for each prototype:

$$\Delta\mu_m = \sum_{b=1}^B \sum_{n=1}^N \alpha_m(\mathbf{x}_{b,n}) \mathbf{x}_{b,n} \quad (9)$$

We apply an Exponential Moving Average (EMA) strategy to perform smooth updates, following the momentum update mechanism in [14],

$$\mu_m^{(t+1)} = (1 - \eta_t) \mu_m^{(t)} + \eta_t \Delta\mu_m, \quad \eta_t = \eta_0 e^{-\kappa t} \quad (10)$$

where η_t denotes the EMA update rate which exponentially decays over training steps:

In practice, to ensure global consistency under distributed training, the writing increments are averaged across all devices to form a synchronized global update.

3.2. Multiscale Prototype Memory

Real-world documents often exhibit multi-scale and hierarchical layout structures and style characteristics. For example, at a coarse scale, a region may appear as a table, while at a finer scale, the same region may correspond to text split by a vertical separator. Such cross-scale hierarchical patterns make it difficult for a single-level memory module to simultaneously capture semantic, structural, and visual details.

Therefore, we introduce multi-scale prototype memory. We build three independent prototype memory modules at different resolutions, each responsible for storing information at a specific scale.

For a visual feature at scale s , denoted $\mathbf{x}^{(s)} \in \mathbb{R}^{B \times N_s \times D_s}$, its corresponding prototype memory module $M^{(s)}$ performs reading and writing via the attention mechanism. The retrieved memory outputs are then downsampled to match the

final encoder output, after which they are concatenated and projected before being fed into the decoder.

$$\tilde{\mathbf{x}} = \text{Proj}[\mathbf{x} \bigoplus_s M^{(s)}(\mathbf{x}^{(s)})] \quad (11)$$

3.3. Objective Function

Prototype memory serves as a compressed representation of spatial layout and stylistic patterns that appear in the training corpus. To ensure that the memory slots learn representative and mutually distinct prototype centers, we encourage each feature token to interact with only a small subset of prototypes during both reading and writing.

Therefore, we introduce an entropy-based sparsity regularization [31]. Let $\alpha_{b,n,m}$ denote the attention coefficient assigned by the n -th encoder token of the b -th sample to the m -th prototype. We define the local entropy loss as:

$$\mathcal{L}_{\text{local_entropy}} = \mathbf{E}_{b,n} \left[- \sum_{m=1}^M \alpha_{b,n,m} \log \alpha_{b,n,m} \right] \quad (12)$$

which encourages each token's attention distribution to be sharp, activating only a small number of prototypes. Further, in order to prevent all tokens from collapsing onto a few prototypes, we introduce a global negative entropy loss that encourages the prototype usage to remain balanced across samples and spatial positions:

$$\mathcal{L}_{\text{global_neg_entropy}} = - \sum_{m=1}^M \mathbf{E}_{b,n} [\alpha_{b,n,m}] \log \mathbf{E}_{b,n} [\alpha_{b,n,m}] \quad (13)$$

The sparsity regularization is defined as the sum of the following two terms:

$$\mathcal{L}_{\text{sparse}} = \mathcal{L}_{\text{local_entropy}} + \mathcal{L}_{\text{global_neg_entropy}} \quad (14)$$

Finally, we combine the document recognition cross-entropy loss with the multiscale sparsity regularization to obtain the objective function:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \sum_s \mathcal{L}_{\text{sparse}}^{(s)} \quad (15)$$

where $\mathcal{L}_{\text{CE}}$ denotes the document recognition cross-entropy loss, and λ is a weighting coefficient.

3.4. Network Architecture

The proposed explicit adaptive multiscale memory module can be seamlessly integrated into a wide range of encoder-decoder architectures. This is because any encoder that produces spatially structured tokens can interact with the memory slots through an attention mechanism. The encoded features can query the memory to retrieve relevant prototype

Table 1. Experiment results on the Fox dataset

Method	Params	Edit Distance↓		F1-score↑		Precision↑		Recall↑		BLEU↑		METEOR↑	
		CN	EN	CN	EN	CN	EN	CN	EN	CN	EN	CN	EN
LLaVA-NeXT [26]	34B	-	0.430	-	0.647	-	0.573	-	0.881	-	0.478	-	0.582
InternVL-ChatV1.5 [6]	26B	0.265	0.393	0.816	0.751	0.784	0.698	0.866	0.917	0.622	0.568	0.717	0.663
Nougat [5]	250B	-	0.255	-	0.745	-	0.720	-	0.809	-	0.665	-	0.761
Vary [36]	7B	0.113	0.092	0.952	0.918	0.961	0.906	0.944	0.956	0.754	0.885	0.873	0.926
Vary-toy [37]	1.8B	0.142	0.082	0.914	0.924	0.928	0.919	0.907	0.938	0.718	0.889	0.832	0.929
Qwen-VL-Plus [1]	>100B	0.121	0.096	0.895	0.931	0.903	0.921	0.890	0.950	0.684	0.893	0.828	0.936
Baseline	0.6B	0.112	0.107	0.944	0.925	0.955	0.922	0.937	0.928	0.753	0.896	0.864	0.905
DREAM (Ours)	0.6B	0.101	0.082	0.964	0.939	0.974	0.933	0.953	0.955	0.767	0.909	0.881	0.942

representations that assist the decoder. During training, the features can also be written back into the memory, allowing the memory slots to gradually form prototype abstractions of the training set.

We apply our method to a large multimodal document understanding model. To further demonstrate the generality of our approach across different text recognition tasks, we additionally integrate our memory module into a handwriting text line recognition model.

3.4.1. Document Recognition Model

Our document recognition model follows an encoder-decoder design, building upon the general framework used in GOT [38]. The encoder adopts the ViT-Det architecture [23], while the decoder is based on the 500M-parameter Qwen model [1]. Between the encoder and decoder, we introduce our explicit multiscale memory module.

To model memory information at different hierarchical levels, we construct three prototype memory modules at three spatial scales. Our encoder outputs visual features at three spatial resolutions, which are used as queries to retrieve responses from the corresponding memory modules.

- $M^{(64)}$: high-level memory capturing semantic-level global layout patterns. The feature tokens correspond to 64×64 pixel patches in the original image;
- $M^{(32)}$: memory modeling mid-level structural patterns.
- $M^{(16)}$: memory representing fine-grained visual styles.

The overall architecture of our DREAM: Document Recognition model with Explicit Adaptive Memory is illustrated in Fig. 2.

3.4.2. Handwriting Recognition Model

Our handwriting text line recognition model is built upon an encoder-decoder framework following [41]. It consists of a ResNet encoder [13] for feature extraction and a Transformer decoder [35] that autoregressively produces the final transcription. The visual features extracted by the ResNet encoder are used as queries to access the prototype memory modules through an attention mechanism.

Unlike square document images, text lines exhibit extreme aspect ratios (e.g., $>12:1$) and variable lengths. Since their large-scale structural patterns often exceed the largest inscribed square, applying standard square-based multiscale memory modules is infeasible.

To address this issue, we introduce an inter-memory self-attention mechanism that enables the prototype memory to model high-level global styles and structural patterns across the entire text line.

4. Experiment Results

In this section, the proposed method is validated on two tasks, i.e. document recognition task on the Fox dataset and DreamDoc, and handwriting recognition task on the SCUT-HCCDoc and SCUT-EPT Chinese handwriting datasets. Ablation experiments are also carried out. Visualization for the memory mechanism are described. Error analysis is also presented.

4.1. Experiment on Document Recognition

4.1.1. Datasets and Evaluation Metrics

Our training data consists of a self-constructed document dataset, DreamDoc, and the Vary-600K training dataset [36].

Our DreamDoc dataset is a bilingual (Chinese-English) document dataset containing 4,800 pages for training and 108 pages for testing, across seven categories: announcements, handwritten notes, slide pages, primary/secondary school textbooks, university textbooks, magazines, and newspapers. The dataset covers a wide range of layout styles including single-column, double-column, complex layouts, and image-text mixed pages making it well suited as training data for our DREAM model. The diverse multi-scale spatial structures and visual styles present in DreamDoc serve as prototypes to be stored by our proposed explicit multiscale memory module.

The Vary-600K dataset is a large-scale PDF document image-text paired dataset released as part of the Vary project.

Table 2. Experiment results on the DreamDoc dataset

Method	Edit Distance↓	F1-score↑	Precision↑	Recall↑	BLEU↑	METEOR↑
GOT [38]	0.473	0.744	0.723	0.794	0.354	0.657
Baseline	0.254	0.868	0.862	0.892	0.731	0.815
DREAM(Ours)	0.248	0.869	0.863	0.893	0.735	0.819

It contains approximately 600,000 document pages, with about 300,000 English pages and 300,000 Chinese pages.

For evaluation, we use the Fox dataset [25] and DreamDoc test set.

The Fox dataset is a document recognizing dataset containing 212 pages, with 112 pages in English and 100 in Chinese. Each page includes more than 1,000 words.

Our DREAM model for document recognition is an pre-OCR free, end-to-end model that directly predicts full-page Markdown, naturally supporting tables.

We evaluate model performance using the following metrics: BLEU [29], METEOR [3], Edit Distance [19], F1-score, Precision and Recall.

4.1.2. Implementation Details

Due to computational resource constraints, we initialize our model from the GOT [38] pretrained checkpoint and fine-tune it in a partial-freezing setting: the visual encoder is kept frozen, while the projection layers, multiscale prototype memory modules, and decoder are trained.

In Eq. 10, η_0 is set to 1×10^{-5} . In Eq. 15, λ is set to 0.1. The number of prototype of the memory module is set to 2048. The AdamW optimizer is used with an initial learning rate of 2×10^{-5} . Training is conducted on eight Huawei Ascend 910B accelerators with batch size set to 2.

4.1.3. Comparison with Other Methods

The comparison with existing methods on the Fox dataset is reported in Table 1. As can be seen, our method achieves the best performance among all compared approaches while using the smallest number of parameters.

We further compare our method with GOT [38] on the DreamDoc test set. The results reported in Table 2, show that our method achieves better performance.

4.1.4. Ablation Study

To validate the effectiveness of the proposed DREAM architecture, we conduct an ablation study on the Fox dataset. We experiment with (i) using only a single-scale prototype memory, (ii) removing the entropy-based sparsity loss $\mathcal{L}_{\text{sparse}}$, and (iii) removing the $\mathcal{L}_{\text{global_neg_entropy}}$ term from the loss (denoted as $+\mathcal{L}_{\text{local_entropy}}$). The results are summarized in Table 3. The findings show that incorporating multiscale adaptive memory and the sparsity loss yields improvements, whereas removing $\mathcal{L}_{\text{global_neg_entropy}}$ leads to a substantial performance drop likely due to attention collapse.

4.1.5. Visualization

To verify that the proposed prototype memory module can capture stable spatial patterns, we visualize its attention coefficients. For each patch feature $\mathbf{x}$ in the image, we compute the attention weights $\{\alpha_m(\mathbf{x})\}$ over all prototypes and find the index of the prototype with the largest response. Thus we find for each patch, which prototype in the memory bank is most strongly activated. The visualization results are shown in Fig. 3. We observe that many cells along table borders share the same dominant prototype response (in purple), while blank regions tend to activate another prototype pattern (in yellow), indicating that the prototype memory have learned different structural patterns.

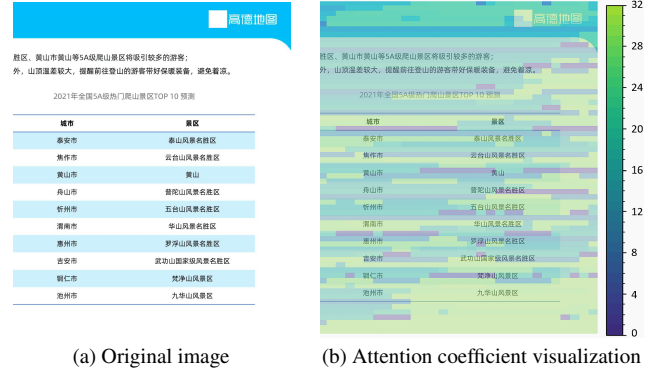


Figure 3. Visualization of attention coefficients for the prototype memory at scale $M^{(32)}$. For clarity, the number of prototypes is set to 32. Each color represent an index of the prototype with the highest attention weight.

4.2. Experiment on Handwriting Recognition

4.2.1. Datasets and Evaluation Metrics

The SCUT-HCCDoc [43] dataset is a large-scale dataset consisting of camera captured Chinese handwriting images obtained by Internet search. There are both horizontal and vertical text images in the dataset. It contains 93,254 training images and 23,389 test images. The SCUT-EPT [45] dataset is collected from examination papers with numerous challenging cases including character erasure, etc. It contains 40,000 training images and 10,000 test images.

For handwriting recognition task, the evaluation metric is Accurate Rate (AR), which is calculated by using edit

Table 3. Ablation study on the Fox dataset

Method	Edit Distance↓		F1-score↑		Precision↑		Recall↑		BLEU↑		METEOR↑	
	CN	EN	CN	EN	CN	EN	CN	EN	CN	EN	CN	EN
Baseline	0.112	0.107	0.944	0.925	0.955	0.922	0.937	0.928	0.753	0.896	0.864	0.905
Single-scale memory + $\mathcal{L}_{\text{sparse}}$	0.126	0.097	0.936	0.941	0.947	0.937	0.929	0.950	0.741	0.915	0.853	0.939
Multiscale memory	0.104	0.096	0.958	0.919	0.967	0.914	0.953	0.936	0.770	0.892	0.882	0.922
Multiscale memory + $\mathcal{L}_{\text{local_entropy}}$	0.136	0.087	0.941	0.926	0.951	0.921	0.932	0.946	0.744	0.891	0.856	0.927
Multiscale memory + $\mathcal{L}_{\text{sparse}}$	0.101	0.082	0.964	0.939	0.974	0.933	0.953	0.955	0.767	0.909	0.881	0.942

Table 4. Experimental results on the SCUT-HCCDoc dataset

Method	AR(%)
SegCTC(2023) [15]	92.08
Li et al.(2023) [22]	92.72
Yao et al. (2024)	92.78
DREAM (Ours)	93.36

Table 5. Experimental results on the SCUT-EPT dataset

Method	AR(%)
Zhu et al.(2018) [45]	75.97
Yan et al.(2021) [40]	76.42
Yao et al (2024) [41]	78.48
DREAM (Ours)	78.82

distance between the recognition results and the ground truth text.

4.2.2. Implementation Details

We use the ADADELTA [42] optimizer with an initial learning rate of 0.3. Training is conducted on four Huawei Ascend 910B accelerators with batch size set to 32.

4.2.3. Comparison with Other Methods

The comparisons of our method and other methods are compared on the SCUT-HCCDoc and SCUT-EPT datasets as shown in Table 4 and Table 5. The experimental results show that, our method has outperformed other reported methods on the two datasets. On the SCUT-HccDoc dataset, our method has achieved the AR of 93.36%, which is better than Yao et al. [41]. On the SCUT-EPT dataset, our method has achieved the AR of 78.82%, also better than Yao et al. [41].

Some examples of recognition results are shown in Figure 4, which demonstrate that our method can adapt challenging situations such as cursive text and noises.

5. Conclusion

We propose DREAM: Document Recognition with Explicit Adaptive Memory, which uses prototype memory to explic-

Input image:	Input image:
Ground-truth: 无法诉诸文字	Ground-truth: 千里马常有
Baseline: 无法诉诸主字	Baseline: 千里马常有
Our method: 无法诉诸文字	Our method: 千里马常有
(a) An example from the SCUT-HccDoc dataset	(b) An example from the SCUT-EPT dataset

Figure 4. Examples of recognition results

itly construct corpus-level memory of multi-scale spatial structures and visual styles in documents. We read from the memory via an attention-based routing and form prototype memories during training. An entropy-based sparsity loss is introduced to encourage fewer prototypes to be activated. We implement a plug-and-play DREAM module and apply it to both a document recognition model and a handwriting text line recognition model. Experimental results show that incorporating the DREAM module leads to significant performance improvements.

The limitations of our approach may include:

- The EMA-based updating strategy for forming prototype memories is slow to adapt, which may hinder rapid adaptation to new data distributions.
- Although the retrieved prototype memories improve performance, it remains unclear how the decoder internally integrates and leverages these retrieved representations.

Our future work will explore a more comprehensive mechanism that integrates short-term and long-term memory, where dynamic clustering is applied to short-term memory to eventually form more robust long-term prototypes.

6. Acknowledgements

This work was supported by Huawei Noah’s Ark Lab. The authors are grateful to Dr. Hailin Hu for useful discussions. The authors thank Yueyun Yang and Jiangze Xiao for assistance with data collection. The authors also thank the anonymous reviewers for their valuable comments that improved the quality of the paper.

References

- [1] Jinze Bai, Shuai Bai, Yunfei Chu, Zeyu Cui, Kai Dang, Xiaodong Deng, Yang Fan, Wenbin Ge, Yu Han, Fei Huang, et al. Qwen technical report. *arXiv preprint arXiv:2309.16609*, 2023. 6
- [2] Ayan Banerjee, Sanket Biswas, Josep Lladós, and Uma-pada Pal. Swindocsegmenter: An end-to-end unified domain adaptive transformer for document instance segmentation. In *International Conference on Document Analysis and Recognition*, pages 307–325. Springer, 2023. 2
- [3] Satanjeev Banerjee and Alon Lavie. METEOR: An automatic metric for mt evaluation with improved correlation with human judgments. In *ACL Workshops*, pages 65–72, 2005. 7
- [4] Manuele Barraco, Sara Sarto, Marcella Cornia, Lorenzo Baraldi, and Rita Cucchiara. With a little help from your own past: Prototypical memory networks for image captioning. In *ICCV*, pages 3021–3031, 2023. 3
- [5] Lukas Blecher, Guillem Cucurull, Thomas Scialom, and Robert Stojnic. Nougat: Neural optical understanding for academic documents. *arXiv preprint arXiv:2308.13418*, 2023. 6
- [6] Zhe Chen, Weiyun Wang, Hao Tian, Shenglong Ye, Zhangwei Gao, Erfei Cui, Wenwen Tong, Kongzhi Hu, Jiapeng Luo, Zheng Ma, et al. How far are we to GPT-4V? closing the gap to commercial multimodal models with open-source suites. *Sci. China Inf. Sci.*, 67(12):220101, 2024. 6
- [7] Zhe Chen, Jiannan Wu, Wenhai Wang, Weijie Su, Guo Chen, Sen Xing, Muyan Zhong, Qinglong Zhang, Xizhou Zhu, Lewei Lu, Bin Li, Ping Luo, Tong Lu, Yu Qiao, and Jifeng Dai. Intern-VL: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. In *CVPR*, pages 24185–24198, 2024. 1, 2
- [8] Cheng Cui, Ting Sun, Suyin Liang, Tingquan Gao, Zelun Zhang, Jiaxuan Liu, Xueqing Wang, Changda Zhou, Hongen Liu, Manhui Lin, et al. PaddleOCR-VL: Boosting multilingual document parsing via a 0.9 b ultra-compact vision-language model. *arXiv preprint arXiv:2510.14528*, 2025. 1, 2
- [9] Jonathan Daume, Jan Kamiński, Andrea GP Schjetnan, Yousef Salimpour, Umair Khan, Michael Kyzar, Chrystal M Reed, William S Anderson, Taufik A Valiante, Adam N Mamelak, et al. Control of working memory by phase-amplitude coupling of human hippocampal neurons. *Nature*, 629(8011):393–401, 2024. 3
- [10] Alex Graves, Greg Wayne, and Ivo Danihelka. Neural turing machines, 2014. 3
- [11] Alex Graves, Greg Wayne, Malcolm Reynolds, Tim Harley, Ivo Danihelka, Agnieszka Grabska-Barwińska, Sergio Gómez Colmenarejo, Edward Grefenstette, Tiago Ramalho, John Agapiou, et al. Hybrid computing using a neural network with dynamic external memory. *Nature*, 538(7626):471–476, 2016. 3
- [12] Tengda Han, Weidi Xie, and Andrew Zisserman. Memory-augmented dense predictive coding for video representation learning. In *ECCV*, 2020. 3
- [13] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, pages 770–778, 2016. 6
- [14] Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. Momentum contrast for unsupervised visual representation learning. In *CVPR*, pages 9729–9738, 2020. 5
- [15] Jiarong Huang, Dezhi Peng, Hongliang Li, Hao Ni, and Lianwen Jin. SegCTC: Offline handwritten Chinese text recognition via better fusion between explicit and implicit segmentation. In *ICDAR*, pages 332–349, 2023. 8
- [16] Marta Huelin Gorris, Masahiro Takigawa, and Daniel Bendor. The role of experience in prioritizing hippocampal replay. *Nat. Commun.*, 14(1):8157, 2023. 3
- [17] Yire Jeong, Hye-Yeon Cho, Mujun Kim, Jung-Pyo Oh, Min Soo Kang, Miran Yoo, Han-Sol Lee, and Jin-Hee Han. Synaptic plasticity-dependent competition rule influences memory formation. *Nat. Commun.*, 12(1):3915, 2021. 3
- [18] Geewook Kim, Teakgyu Hong, Moonbin Yim, JeongYeon Nam, Jinyoung Park, Jinyeong Yim, Wonseok Hwang, Sangdoo Yun, Dongyoon Han, and Seunghyun Park. OCR-free document understanding transformer. In *ECCV*, pages 498–517, 2022. 1, 2
- [19] V.I. Lvenshtcin. Binary coors capable or ‘correcting deletions, insertions, and reversals. In *Soviet physics-doklady*, 1966. 7
- [20] Minhyeok Lee, Suhwan Cho, Seunghoon Lee, Chaewon Park, and Sangyoun Lee. Unsupervised video object segmentation via prototype memory network. In *WACV*, pages 5924–5934, 2023. 3
- [21] Tianqin Li, Zijie Li, Andrew Luo, Harold Rockwell, Amir Barati Farimani, and Tai Sing Lee. Prototype memory and attention mechanisms for few shot image generation. In *ICLR*, 2022. 3
- [22] Tao Li, Shilian Wu, and Zengfu Wang. Mask guided selective context decoding for handwritten Chinese text recognition. In *ICASSP*, pages 1–5, 2023. 8
- [23] Yanghao Li, Hanzhi Mao, Ross Girshick, and Kaiming He. Exploring plain vision transformer backbones for object detection. In *ECCV*, pages 280–296. Springer, 2022. 6
- [24] Zhang Li, Biao Yang, Qiang Liu, Zhiyin Ma, Shuo Zhang, Jingxu Yang, Yabo Sun, Yuliang Liu, and Xiang Bai. Monkey: Image resolution and text label are important things for large multi-modal models. In *CVPR*, pages 26763–26773, 2024. 1, 2
- [25] Chenglong Liu, Haoran Wei, Jinyue Chen, Lingyu Kong, Zheng Ge, Zining Zhu, Liang Zhao, Jianjian Sun, Chunrui Han, and Xiangyu Zhang. Focus anywhere for fine-grained multi-page document understanding. *arXiv preprint arXiv:2405.14295*, 2024. 2, 7
- [26] Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan Zhang, Sheng Shen, and Yong Jae Lee. Llava-next: Improved reasoning, OCR, and world knowledge, 2024. 6
- [27] Haoyu Lu, Wen Liu, Bo Zhang, Bingxuan Wang, Kai Dong, Bo Liu, Jingxiang Sun, Tongzheng Ren, Zhuoshu Li, Hao Yang, Yaofeng Sun, Chengqi Deng, Hanwei Xu, Zhenda Xie, and Chong Ruan. DeepSeek-VL: Towards real-world vision-language understanding, 2024. 1, 2
- [28] Randall C O’Reilly, Rajan Bhattacharyya, Michael D Howard, and Nicholas Ketz. Complementary learning systems. *Cognitive science*, 38(6):1229–1248, 2014. 3

- [29] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic evaluation of machine translation. In ACL, pages 311–318, 2002. [7](#)
- [30] Birgit Pfitzmann, Christoph Auer, Michele Dolfi, Ahmed S Nassar, and Peter Staar. Doclaynet: A large human-annotated dataset for document-layout segmentation. In Proceedings of the 28th ACM SIGKDD conference on knowledge discovery and data mining, pages 3743–3751, 2022. [2](#)
- [31] Claude E Shannon. A mathematical theory of communication. BSTJ, 27(3):379–423, 1948. [5](#)
- [32] Zejiang Shen, Ruochen Zhang, Melissa Dell, Benjamin Charles Germain Lee, Jacob Carlson, and Weining Li. Lay-outparser: A unified toolkit for deep learning based document image analysis. In International Conference on Document Analysis and Recognition, pages 131–146. Springer, 2021. [2](#)
- [33] Enxin Song, Wenhao Chai, Tian Ye, Jenq-Neng Hwang, Xi Li, and Gaoang Wang. MovieChat+: Question-aware sparse memory for long video question answering. arXiv preprint arXiv:2404.17176, 2024. [3](#)
- [34] Jochen Triesch, Anh Duong Vo, and Anne-Sophie Hafner. Competition for synaptic building blocks shapes synaptic plasticity. Elife, 7:e37836, 2018. [3](#)
- [35] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. NeurIPS, 30, 2017. [2](#), [4](#), [6](#)
- [36] Haoran Wei, Lingyu Kong, Jinyue Chen, Liang Zhao, Zheng Ge, Jinrong Yang, Jianjian Sun, Chunrui Han, and Xiangyu Zhang. Vary: Scaling up the vision vocabulary for large vision-language model. In ECCV, pages 408–424. Springer, 2024. [2](#), [6](#)
- [37] Haoran Wei, Lingyu Kong, Jinyue Chen, Liang Zhao, Zheng Ge, En Yu, Jianjian Sun, Chunrui Han, and Xiangyu Zhang. Small language model meets with reinforced vision vocabulary. arXiv preprint arXiv:2401.12503, 2024. [6](#)
- [38] Haoran Wei, Chenglong Liu, Jinyue Chen, Jia Wang, Lingyu Kong, Yanming Xu, Zheng Ge, Liang Zhao, Jianjian Sun, Yuang Peng, et al. General OCR theory: Towards OCR-2.0 via a unified end-to-end model. arXiv preprint arXiv:2409.01704, 2024. [1](#), [2](#), [6](#), [7](#)
- [39] Yiheng Xu, Minghao Li, Lei Cui, Shaohan Huang, Furu Wei, and Ming Zhou. LayoutLM: Pre-training of text and layout for document image understanding. In KDD, pages 1192–1200, 2020. [1](#), [2](#)
- [40] Shi Yan, Jin-Wen Wu, Fei Yin, and Cheng-Lin Liu. Recognizing handwritten Chinese texts with insertion and swapping using a structural attention network. In ICDAR, pages 557–571, 2021. [8](#)
- [41] Gang Yao, Ning Ding, Tianqi Zhao, Kemeng Zhao, Pei Tang, Yao Tao, and Liangrui Peng. Visual prompt learning for chinese handwriting recognition. In ICDAR, pages 119–133, 2024. [6](#), [8](#)
- [42] Matthew D Zeiler. Adadelata: An adaptive learning rate method. arXiv preprint arXiv:1212.5701, 2012. [8](#)
- [43] Hesuo Zhang, Lingyu Liang, and Lianwen Jin. SCUT-HCCDoc: A new benchmark dataset of handwritten Chinese text in unconstrained camera-captured documents. PR, 108: 107559, 2020. [2](#), [7](#)
- [44] Xu Zhong, Jianbin Tang, and Antonio Jimeno Yepes. Publaynet: largest dataset ever for document layout analysis. In 2019 International conference on document analysis and recognition (ICDAR), pages 1015–1022. IEEE, 2019. [2](#)
- [45] Yanzhi Zhu, Zecheng Xie, Lianwen Jin, Xiaoxue Chen, Yaoxiong Huang, and Ming Zhang. SCUT-EPT: New dataset and benchmark for offline Chinese text recognition in examination paper. IEEE Access, 7:370–382, 2018. [2](#), [7](#), [8](#)